

Automated Essay Scoring of College Application Essays with LLMs

Jinwoo Kim, Stephen Slade (advisor)

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1 Introduction

The COVID-19 pandemic catalyzed an unprecedented shift toward online education, making digital learning environments an inevitable part of modern academia. One sector that has seen significant adaptation is the college admissions consulting industry, a multi-billion-dollar business that traditionally relies on human reviewers to assess and provide feedback on student application essays. However, manual revision of student essays is time-consuming, encouraging consulting firms to explore tools that can expedite the process.

2 Background

The education industry has traditionally relied on human reviewers to evaluate and provide feedback on student application essays, a process that is both costly and labor-intensive. As a result, there has been significant interest in Automated Essay Scoring (AES) as a potential solution to expedite essay evaluation. Despite inconsistencies in human grading, AES models have demonstrated a strong ability to match human scores [1].

While numerous AES models have been developed, recent works like Ruijie et al. [2] and Xiao et al. [3] highlight the power of Large Language Models (LLMs) for this task, which exhibit notable consistency, generalizability, and explainability. However, while LLMs have made significant strides, their accuracy—particularly within the domain of college application essays—still has room for improvement.

2.1 Limitations with AES and LLMs

Existing AES models and LLMs are designed for generability, as they must evaluate essays across diverse domains (e.g., argumentative vs. narrative writing). As a result, they prioritize consistency over domain-specific accuracy. While LLMs are highly capable, they lack fine-tuning for college application essays,

unlike dedicated AES models. Enhancing LLMs through fine-tuning and incorporating key factors identified by college admissions consulting firms presents a significant opportunity for improvement.

2.2 Novelties

We aim to address these limitations through two key approaches:

1. **Domain-Specific Understanding** – Incorporating factors identified by college admissions consulting firms, such as PS ratio, VSPICE, and specificity, into the evaluation criteria.
2. **Fine-Tuning on a Specialized Dataset** – Fine-tuning LLMs on a dataset of 4,000 college application essays, which includes human-assigned scores and feedback, to enhance the model’s ability to accurately assess writing quality.

2.3 Key Terms

Narrowing the domain of essays allows us to focus on key factors prioritized by professional college consulting firms, improving the model’s accuracy and relevance. According to the consulting firm ElevatEd, some primary factors to consider are:

1. **PS Ratio** - College consulting firms (e.g., College Essay Guy, ElevatEd, and Next Admit) stress the importance of a well-balanced problem-solution (PS) ratio in essays. The ideal distribution allocates 20-40% to the problem, 40-60% to the solution, with the remainder reserved for reflection.
2. **SEEPS** - SEEPS is a framework that ensures an essay contains a personal voice and emotion.
3. **VSPICE** - VSPICE stands for Vulnerability, Selflessness, Perseverance, Initiative, and Expression. ElevatEd’s rubric encourages students to incorporate as many of these elements as possible in their essays to create a well-rounded narrative.

3 Project Description

This project aims to develop a model that consistently and accurately evaluates college application essays. The first phase involves using PyTorch and OpenAI packages to fine-tune LLMs on a dataset of 4,000 college application essays. In this dataset, each essay is accompanied by feedback and a score assigned by professional college consultants, with scores ranging from 0 to 9 in 0.5-point increments.

In the second phase, the model will predict scores for key factors such as the PS ratio, SEEPS, and VSPICE, then weigh these scores to generate a final

evaluation. This structured framework ensures that industry-backed criteria are systematically incorporated into the assessment process.

Finally, the final report will include a comparison of the model’s accuracy against existing LLMs to evaluate its effectiveness in scoring college application essays.

4 Proposed Timeline

Week of	Task
Feb 2, 9	Literature review and data preprocessing
Feb 16, 23	Fine-tune OpenAI LLM on data
Mar 2, 9, 16	Train model to predict PS Ratio, SEEPS, and VSPICE scores
Mar 23	Try linear regression and neural networks on the predicted scores to generate a final evaluation
Mar 30, Apr 6	Compare model against baselines (e.g. ChatGPT, Claude) and build front-end for presentation
Apr 13	Draft final report and present preliminary results to advisor
Apr 20	Revise report based on feedback and prepare poster for presentation
Apr 27	Finalize submission and complete presentation materials

5 Deliverables

1. Final report of methodology and results.
2. Website with access to the final evaluation model.
3. Comparative analysis of the fine-tuned model against baseline models (ChatGPT, Claude).
4. Final presentation and project poster.

References

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